

Pavan Kumar

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EDUCATION

Jaypee University of Information Technology

June 2024 – June 2028

Bachelor of Technology in Computer Science and Engineering

Current CGPA: 8.5

SKILLS

AI/Machine Learning	LLM, Computer Vision, Deep Learning, NLP, Data Preprocessing
Languages & Libraries	Python (PyTorch, TensorFlow, Pandas, NumPy, Scikit-learn), SQL, C++
Developer Tools	Git, GitHub, Linux/Bash, REST APIs, Ollama, LaTeX, UNIX Shell
Soft Skills	Critical Thinking, Problem-solving, Adaptability, Communication, Curiosity

EXPERIENCE

Signal Processing Research Trainee

March 2026 - Present

- Conducting research under faculty supervision on signal processing pipelines and waveform analysis.
- Engineered preprocessing and filtering workflows across 5+ signal processing concepts.
- Applied time-domain and frequency-domain analysis techniques to biological signal datasets.
- Gained exposure to EEG signal analysis workflows, brainwave interpretation, and biological signal processing concepts.

PROJECTS

Crop Wise

[GitHub Link](#)

- Built an ML-based crop yield prediction system using 10+ parameters with ~90% accuracy.
- Developed a FastAPI backend with 5+ endpoints for real-time crop recommendations.
- Implemented crop & fertilizer optimization for 7+ varieties, reducing resource wastage by ~20%.
- Proposed a multilingual LLM chatbot with voice support for rural farmers.

Cypher

[GitHub Link](#)

- Built an AI-powered IDS trained on 1.5M+ flows from CICIDS2017 dataset, detecting multi-vector cyber threats in <100ms with 99.2% accuracy, reducing incident response time by ~65%.
- Analyzed 78 network features with 7 engineered metrics (Packet Rate, SYN/ACK Ratio, Bytes per Second) to capture subtle traffic anomalies.
- Designed adaptive ML pipeline to classify unknown threats via similarity-based pattern recognition.
- Implemented a REST-API layer supporting both batch analysis, real-time traffic stream classification.

ACHIEVEMENTS

- **2nd Runner Up** – CyberAIHackathon, University of Derby
- **1st Place** – Capture The Flag (CTF), Cybersecurity Alliance, JUIT
- **Runner Up** – Hacking-Hills Hackathon, IIC